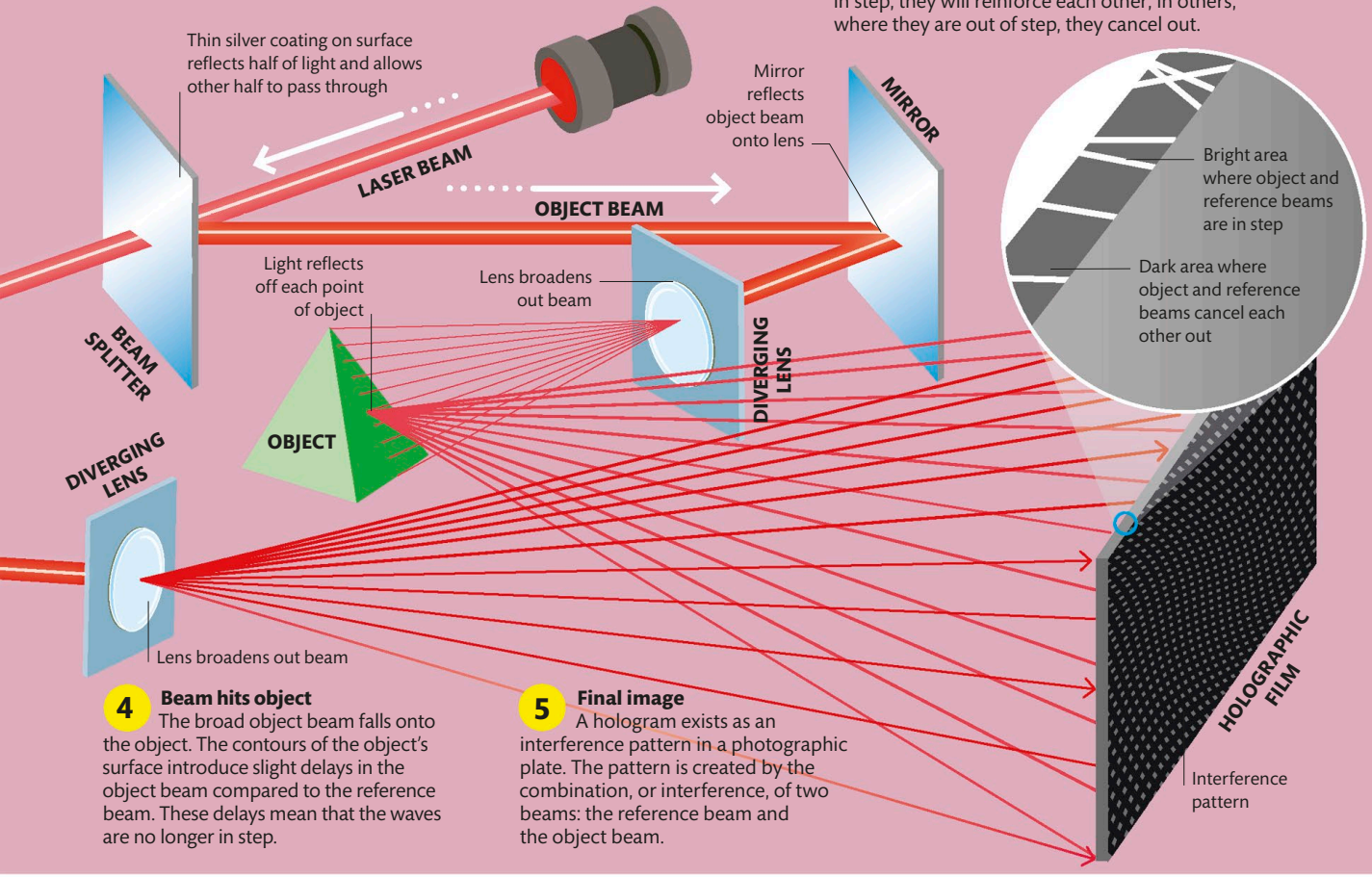
2 Object beam formed
Light that is reflected off the beam splitter forms the object beam. A mirror deflects it toward the object, but first it passes through a lens that broadens out the beam. This is called a diverging lens.

1 Laser fires beam
The light from the laser emerges as a thin beam of coherent light waves. This means that they are all the same wavelength and are all in step with each other.

Holographic plate
Light waves that have reflected off the object's surface will be out of step with the waves of the reference beam. When the two light waves meet inside the photographic plate, they combine, or interfere. In some places, where the waves are still in step, they will reinforce each other; in others, where they are out of step, they cancel out.



4 Beam hits object
The broad object beam falls onto the object. The contours of the object's surface introduce slight delays in the object beam compared to the reference beam. These delays mean that the waves are no longer in step.

5 Final image
A hologram exists as an interference pattern in a photographic plate. The pattern is created by the combination, or interference, of two beams: the reference beam and the object beam.

Viewing a transmission hologram
When light rays bounce off the interference pattern inside the photographic film of a transmission hologram, they recreate the pattern of light that bounced off the object. As a result, they form an image of the object behind the photographic plate. This virtual image has depth and can be viewed from any angle.

